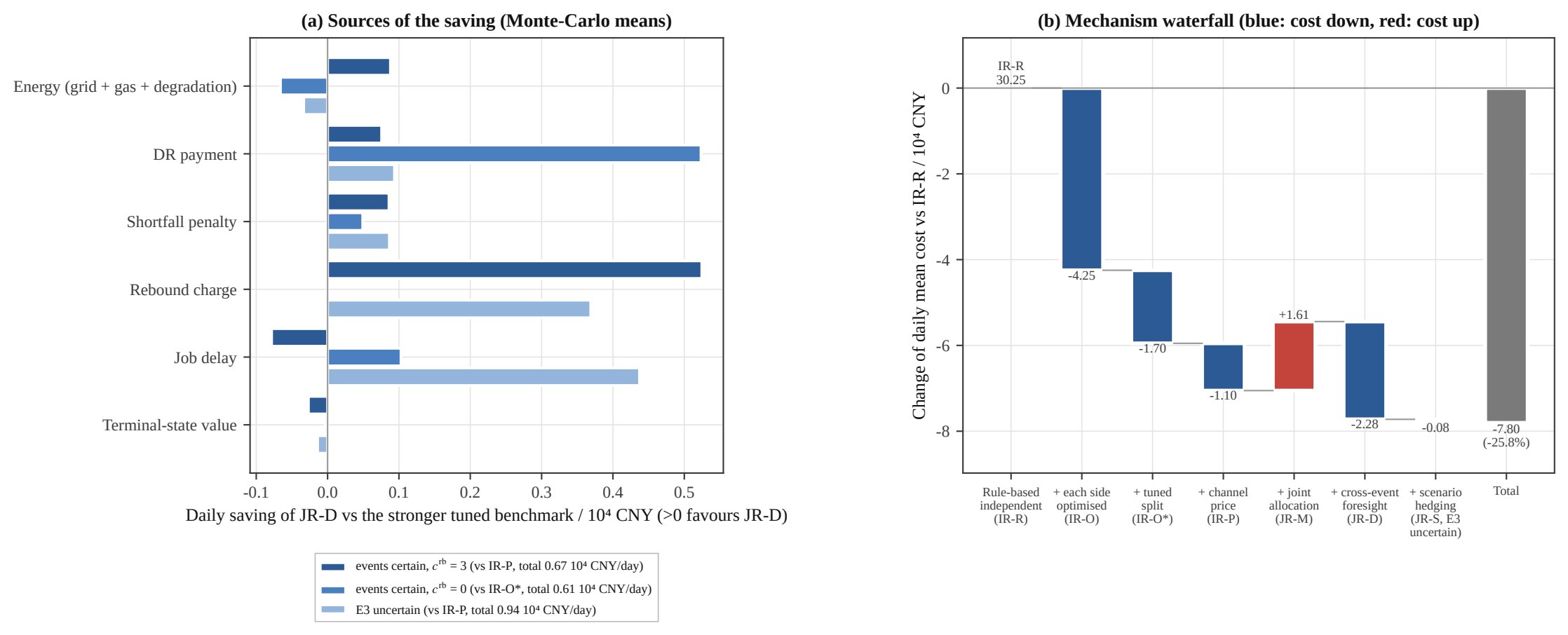
(a) Sources of the saving (Monte-Carlo means)
Energy (grid + gas + degradation)
DR payment
Shortfall penalty
Rebound charge
Job delay
Terminal-state value
-0.1
0.0
0.1
0.2
0.3
0.4
0.5
Daily saving of JR-D vs the stronger tuned benchmark / 10⁴ CNY (>0 favours JR-D)
events certain, c^rb = 3 (vs IR-P, total 0.67 10⁴ CNY/day)
events certain, c^rb = 0 (vs IR-O*, total 0.61 10⁴ CNY/day)
E3 uncertain (vs IR-P, total 0.94 10⁴ CNY/day)
(b) Mechanism waterfall (blue: cost down, red: cost up)
Change of daily mean cost vs IR-R / 10⁴ CNY
0
-2
-4
-6
-8
IR-R
30.25
-4.25
-1.70
-1.10
+1.61
-2.28
-0.08
-7.80
(-25.8%)
Rule-based independent (IR-R)
+ each side optimised (IR-O)
+ tuned split (IR-O*)
+ channel price (IR-P)
+ joint allocation (JR-M)
+ cross-event foresight (JR-D)
+ scenario hedging (JR-S, E3 uncertain)
Total